Aurick Anwar

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Education

McMaster University — Hamilton, Canada
Bachelor of Engineering: Engineering 1 student — September 2025 – June 2030

Work Experience

Founding Engineer — Toronto, Ontario
Magnified Systems — *February 2026 - Present*

- Developing a mounted device that can be attached to a helmet and is used to detect and quantify impact severity in real time.
- Built and trained a PyTorch regression model that predicted an impact severity score from (1-100) at a **96.4%** accuracy off sample crash data (CSV file).
- Using a 6-axis IMU sensor to measure linear and angular acceleration. I plan on finishing a prototype model by the end of June.

Assistant Instructor — Toronto, Ontario
The STEAM Project — June 2025 – August 2025

- Guided ~**10** students per session in hands-on woodworking and robotics projects, safely supervising tools such as scroll saws with **zero safety incidents.**
- Co-developed and improved a Battle Bots curriculum, teaching CAD design and basic electronic platforms to **50+ students** improving project completion rates by ~**40%**.
- Assisted in packaging and delivering **100+** project kits across multiple locations, ensuring timely distribution.

CMO and Co-Founder — Toronto, Ontario
Orbitview — December 2023 – June 2025

- Led a group of **15-20** young high school developers to design and build a scalable and digital platform for more professional content creation.
- Created documentation and guides for AI resume templates and fixes.
- Planned technical events and collaborated with multiple AI startups in Toronto to host webinars. Advertised on all social media platforms by creating posts to attract over **200** people to our webinar, and our revenue exceeded the initial cost by **5x**.

Personal Projects

AI Smart Computer w/ Hand Gesture Control, *MediaPipe, PyAutoGUI, OpenCV*

- Built a real-time gesture-controlled interface, achieving ~**30 FPS** using MediaPipe and OpenCV for touchless computer control.
- Implemented **6+ gesture mappings** with 21-point hand tracking, enabling click, cursor, mute, and volume control via distance thresholds.
- Reduced unintended inputs by ~**60%** using gesture filtering and cooldown logic, achieving **<100 ms** latency.

Basketball Shot Predictor, *OpenCV, YOLOv11, PyTorch*

- Built a real-time basketball shot prediction system using YOLOv11 and OpenCV, processing ~**30 FPS** video and tracking ball positions across **50+ frames** per shot.
- Created and wrote to a dataset with **9+ features** per frame(speed, distance to rim, position of ball) generating **100+** labeled data points across multiple videos for training.
- Designed and trained a PyTorch neural network, using BCE loss and Adam Optimization, learning non-linear relationships between ball trajectory features and shot outcomes, achieving ~**80-85%** prediction accuracy.

Google Home Replica, *Python, Google Cloud, Text-to-speech*

- Built a Google Home/Alexa replica using **Google Cloud APIs** and **Text-to-speech**, processing user queries with **<3s** response time.
- Implemented **4+** command types (math calculations, date queries, music, flipping a coin), enabling real-time voice interaction.

Technical Skills

- Programming Skills: Python, C++
- Libraries: PyTorch, OpenCV, MediaPipe, PyAutoGUI, YOLOv11
- 3D Design: AutoCAD, Fusion360, Inventor
- Electronics and Microcontrollers: Arduino, Raspberry PI, ESP32
- Microsoft Office: Excel, Word, PowerPoint